



DSA Class Notes



Day 1: Intro to Data Structures & Algorithms

by Akshit Sharma



What we'll learn today

- 1) What is a Data Structure?
- 2) What is an Algorithm?
- 3) Types of Data Structures
- 4) Linear vs Non-Linear DS
- 5) Primitive vs Non-Primitive DS

1. What's a Data Structure, really?

It's just a WAY of arranging your data so that using it later is easy and fast.

Think of it like different containers for different jobs →

- ✎ A queue at a ticket window → first come, first served
- ✎ A pile of plates → you take the top one first
- ✎ A family tree → parents connect to children

Each of these is a different “shape” of organizing things — that's exactly what a data structure is, but for data inside a computer!

2. What's an Algorithm?

A step-by-step recipe to solve a problem. Just like a cooking recipe — clear steps, in order, that end with a result.

Example: Find the biggest of 3 numbers

Step 1: Look at a, b, c

Step 2: If a is bigger than both → a is the answer

Step 3: Else if b is bigger than c → b is the answer

Step 4: Else → c is the answer

3. Types of Data Structures — the big picture



Fig 1 — Data Structures split into Primitive & Non-Primitive; Non-Primitive further splits into Linear & Non-Linear

4. Linear vs Non-Linear — what's the difference?

► Linear DS: elements sit one after another

Only ONE way to walk through them — start to end, like a straight line.

1) Array -

10	
----	--

[0]

Fig 2 — Array, Stack, Queue and Linked List are all Linear Data Structures

► **Non-Linear DS: elements branch out / connect**

Elements can connect to MORE THAN ONE other element — no single straight path.

1) Tree —

Fig 3 — Tree (hierarchy) and Graph (network) are Non-Linear Data Structures

Linear DS	Non-Linear DS
One element after another	Elements branch / connect to many
Single level	Multiple levels
One pass to visit everything	Needs BFS / DFS / Inorder etc.
Array, Stack, Queue, Linked List	Tree, Graph

5. Primitive vs Non-Primitive DS

Primitive = the basic building blocks. Non-Primitive = things BUILT using those blocks.

Primitive DS	Non-Primitive DS
int, char, float, boolean...	Array, Linked List, Stack, Queue...
Stores ONE value	Stores a COLLECTION of values
Fixed size	Size can grow/shrink

U0001F4DD Quick Recap

Data Structure = way to organize data.

Algorithm = step-by-step way to solve a problem.

Primitive (basic types) vs Non-Primitive (collections).

Linear (sequential: Array, Stack, Queue, LL) vs Non-Linear (branching: Tree, Graph).

U0001F3AF Try this before next class

Write down 3 real-life examples of Linear DS and 3 of Non-Linear DS (don't repeat the ones from class!). Bring them to the next session — we'll discuss a few.